

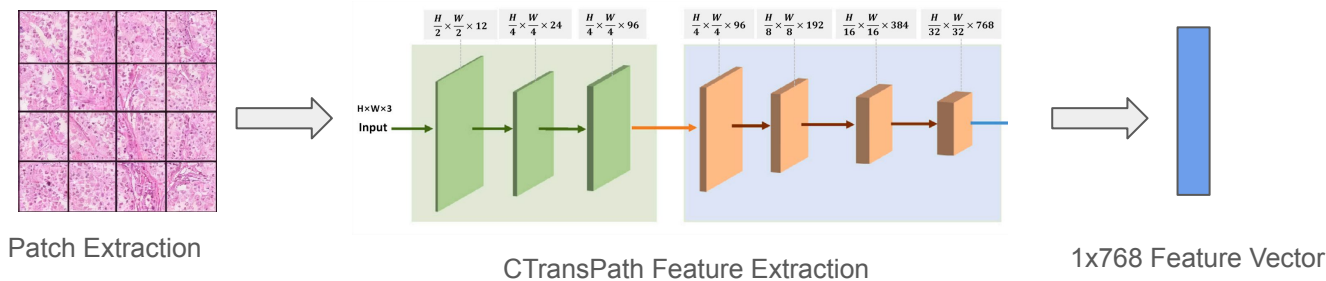


Domain Shift Quantification

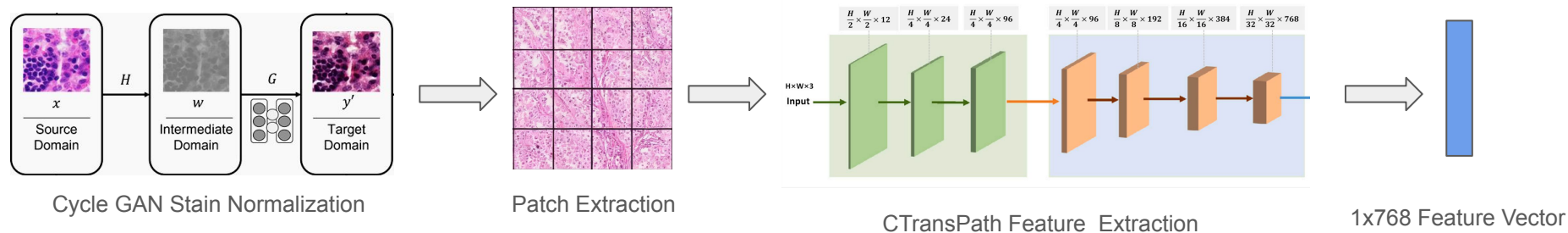
Yang Cao | Lillian Lam | Naina Misra | 02/19/2025

Recap

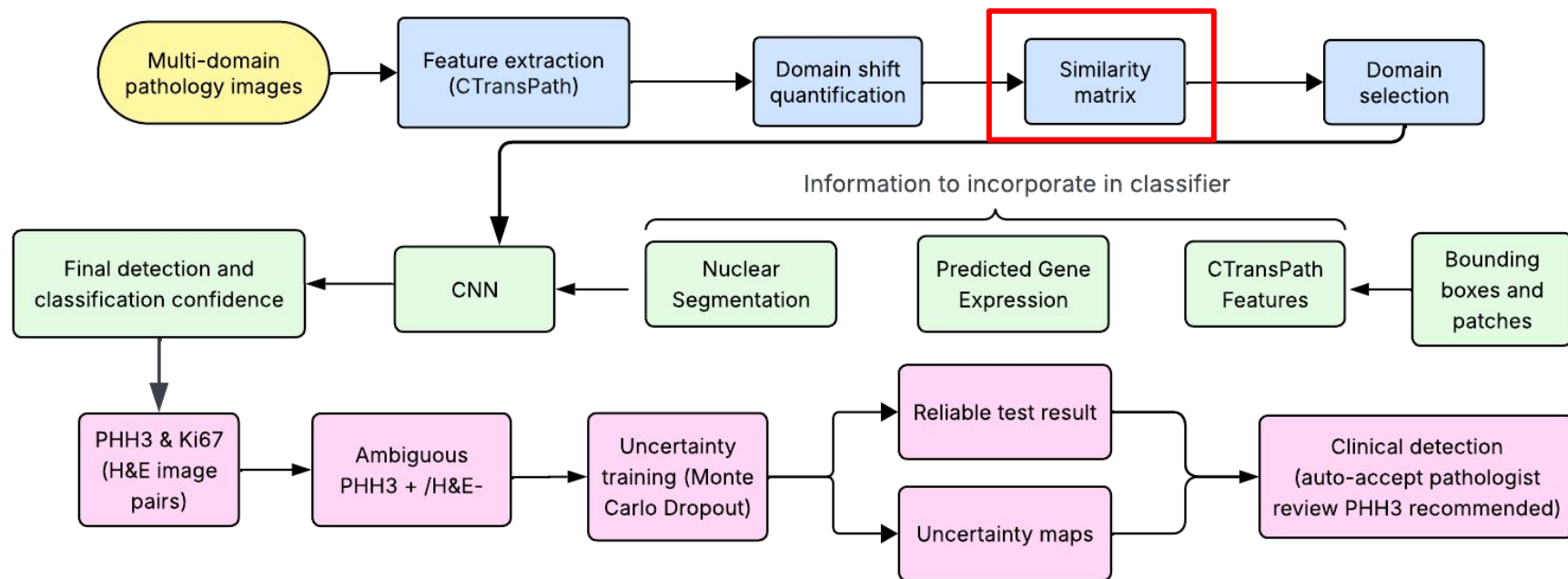
CTransPath Feature Extraction



Stain Normalized CTransPath Feature Extraction



Pipeline



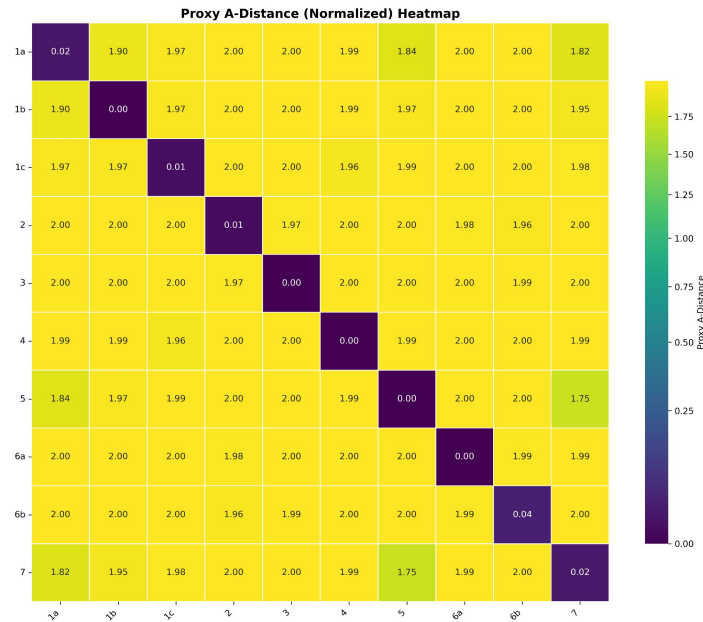
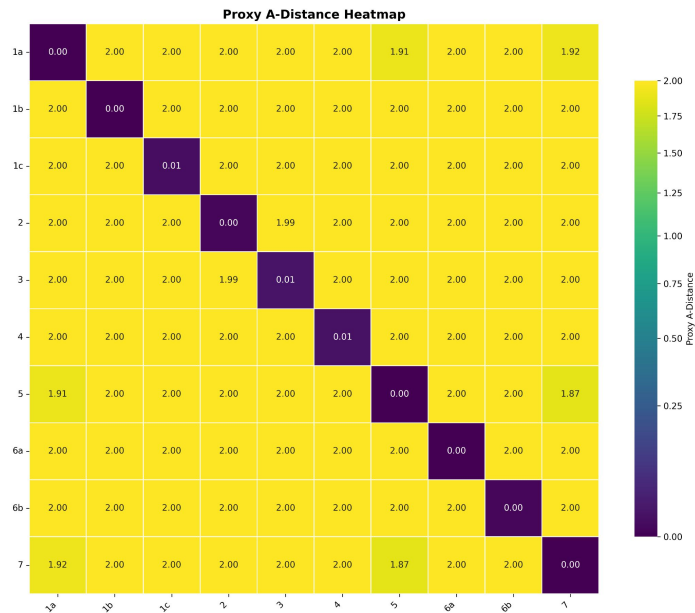
Proxy A-Distance

Proxy A-distance is an empirical metric used to quantify the distributional discrepancy between two domains.

$$\hat{d}_A = 2(1 - 2\epsilon)$$

- ϵ (**epsilon**) is the generalization error (on the test set) of a binary classifier (e.g., Linear SVM) trained to distinguish source samples from target samples.

Proxy A-Distance

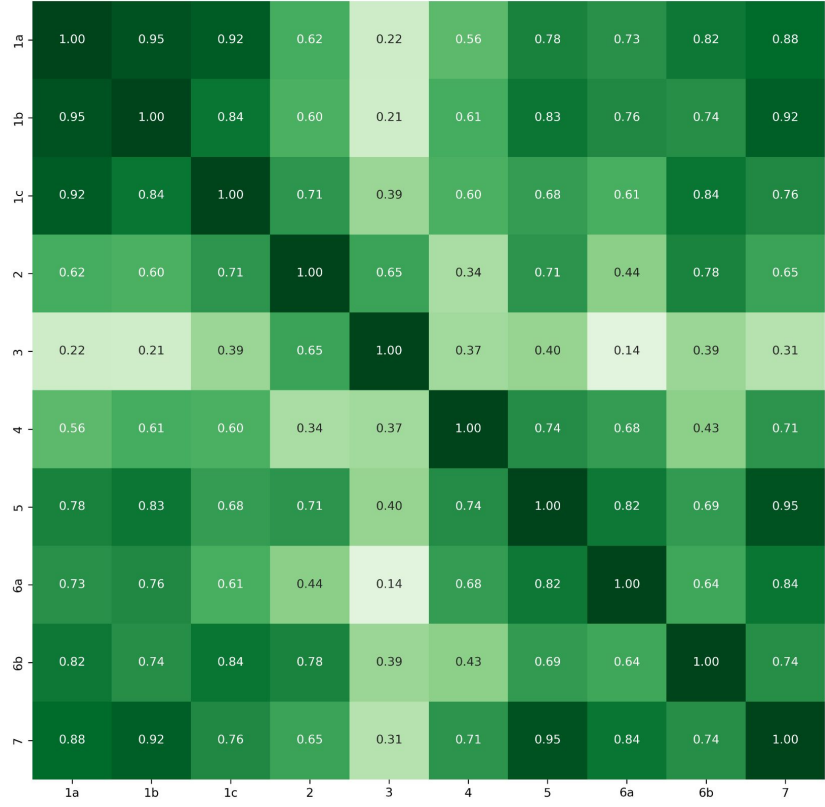


If $\epsilon \approx 0$ (Easy to classify): The distance is **High** (≈ 2.0), indicating severe domain shift.

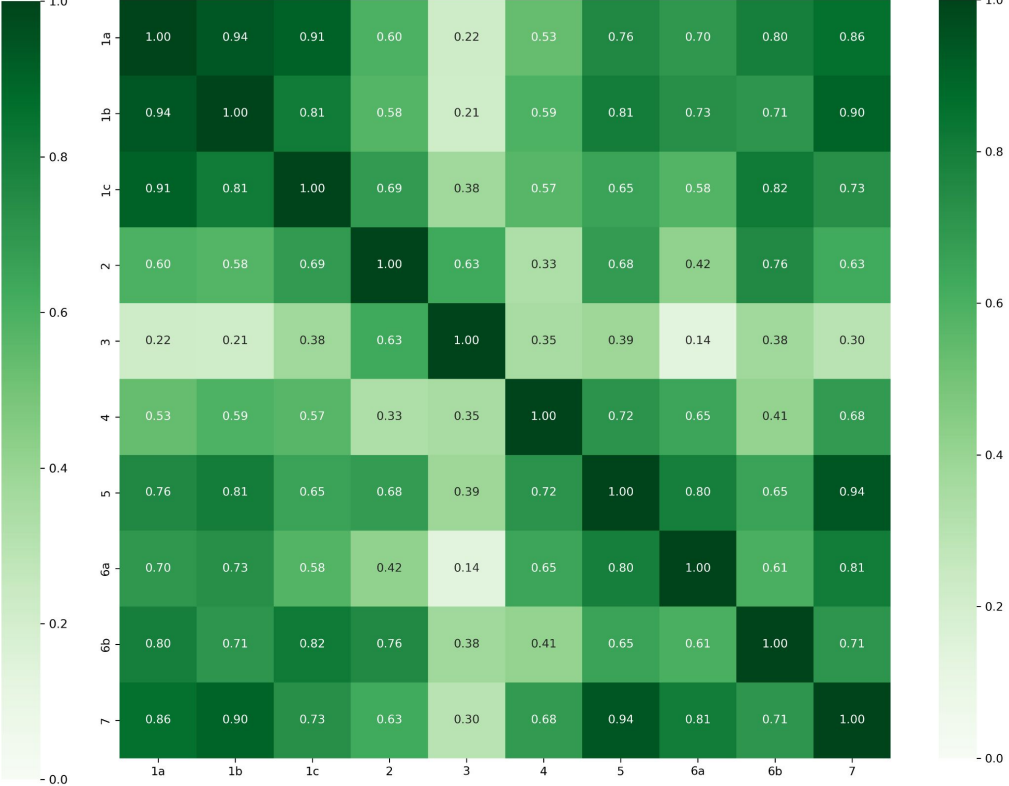
If $\epsilon \approx 0.5$ (Random guessing): The distance is **Low** (≈ 0.0), indicating the domains are well-aligned and indistinguishable.

Similarity Matrix

Final Similarity Matrix
(MMD=0.5, Wass=0.3, CORAL=0.2)



Final Similarity Matrix
(MMD=0.4, Wass=0.4, CORAL=0.2)



Similarity Matrix(WSI)

	1	2	3	4	5	8	9	10	11	12	13	16	17	19	20
1	1	0.981915931	0.887281666	0.943054379	0.980033562	0.936428354	0.974036134	0.943155593	0.953864370	0.961369881	0.977767098	0.967407787	0.959184809	0.934514723	0.946682665
2	0.981915931	1	0.869459779	0.953416044	0.974465209	0.935305618	0.946785309	0.954360036	0.955216366	0.961135995	0.969856411	0.959905845	0.963673413	0.949092814	0.965084374
3	0.887281666	0.869459779	1	0.838362011	0.852569756	0.931687862	0.895235377	0.877015043	0.903360713	0.853518807	0.909670227	0.942626973	0.801544252	0.797844548	0.840622293
4	0.943054379	0.953416044	0.838362011	1	0.953760637	0.923041367	0.913581665	0.956033558	0.902874501	0.952039236	0.930158590	0.921596340	0.940315061	0.951525802	0.965764027
5	0.980033562	0.974465209	0.852569756	0.953760637	1	0.940918040	0.940485953	0.944656746	0.936276222	0.963573912	0.965871941	0.938498415	0.976403488	0.956623213	0.965305788
8	0.936428354	0.935305618	0.931687862	0.923041367	0.940918040	1	0.918465511	0.951486932	0.927945994	0.914860850	0.955845604	0.944977149	0.899581401	0.905699264	0.945525267
9	0.974036134	0.946785309	0.895235377	0.913581665	0.940485953	0.918465511	1	0.914956324	0.953352776	0.949735949	0.967602457	0.971352973	0.926885287	0.892953357	0.903320695
10	0.943155593	0.954360036	0.877015043	0.956033558	0.944656746	0.951486932	0.914956324	1	0.913954056	0.942991220	0.944775745	0.941665015	0.936895761	0.930424056	0.973321455
11	0.953864370	0.955216366	0.903360713	0.902874501	0.936276222	0.927945994	0.953352776	0.913954056	1	0.925039770	0.969801875	0.975251852	0.923228351	0.902689398	0.920324577
12	0.961369881	0.961135995	0.853518807	0.952039236	0.963573912	0.914860850	0.949735949	0.942991220	0.925039770	1	0.946238074	0.945328376	0.961335177	0.962946508	0.955931521
13	0.977767098	0.969856411	0.909670227	0.930158590	0.965871941	0.955845604	0.967602457	0.944775745	0.969801875	0.946238074	1	0.970705466	0.946672897	0.928081628	0.942105375
16	0.967407787	0.959905845	0.942626973	0.921596340	0.938498415	0.944977149	0.971352973	0.941665015	0.975251852	0.945328376	0.970705466	1	0.917756592	0.899568327	0.924852345
17	0.959184809	0.963673413	0.801544252	0.940315061	0.976403488	0.899581401	0.926885287	0.936895761	0.923228351	0.961335177	0.946672897	0.917756592	1	0.951723987	0.958794345
19	0.934514723	0.949092814	0.797844548	0.951525802	0.956623213	0.905699264	0.892953357	0.930424056	0.902689398	0.962946508	0.928081628	0.899568327	0.951723987	1	0.961515474
20	0.946682669	0.965084374	0.840622293	0.965764027	0.965305788	0.945525267	0.903320695	0.973321455	0.920324577	0.955931521	0.942105375	0.924852349	0.958794349	0.961515474	1

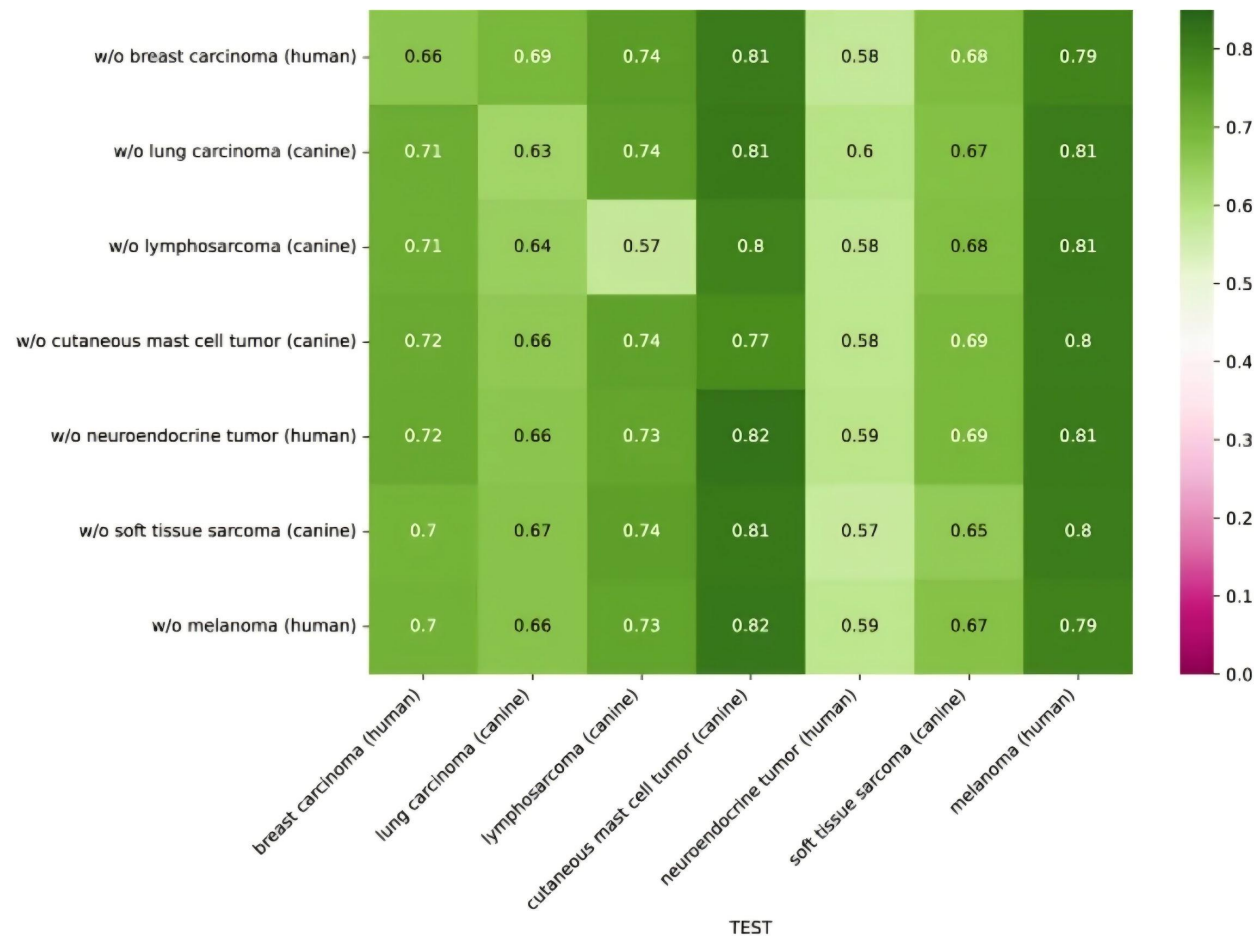
Summary

- We constructed a robust WSI-level Similarity Matrix by fusing three complementary metrics- MMD, CORAL, and Wasserstein- to comprehensively quantify distributional shifts across slides.

Next Time

- Started training with R-CNN with ResNet-50 backbone
- Compare accuracy and F1 score to the ranked WSI images
- Train on only Hematoxylin part of the WSI
- Try a different H&E to Spatial Transcriptomics

Thank you!



Domain	No. Cases	Tumor Type	Origin	Species	Scanner	Resolution
1a	50	breast carcinoma	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23 \frac{\mu m}{px}$
1b	50	breast carcinoma	UMC Utrecht	human	Hamamatsu S360 (0.5 N/A)	$0.23 \frac{\mu m}{px}$
1c	50	breast carcinoma	UMC Utrecht	human	Leica ScanScope CS2	$0.25 \frac{\mu m}{px}$
2	44	lung carcinoma	VMU Vienna	canine	3DHistech Pannoramic Scan II	$0.25 \frac{\mu m}{px}$
3	55	lymphosarcoma	VMU Vienna	canine	3DHistech Pannoramic Scan II	$0.25 \frac{\mu m}{px}$
4	50	cutaneous mast cell tumor	FU Berlin	canine	Aperio ScanScope CS2	$0.25 \frac{\mu m}{px}$
5	55	neuroendocrine tumor	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23 \frac{\mu m}{px}$
6a	85	soft tissue sarcoma	AMC New York	canine	3DHistech Pannoramic Scan II	$0.25 \frac{\mu m}{px}$
6b	15	soft tissue sarcoma	VMU Vienna	canine	3DHistech Pannoramic Scan II	$0.25 \frac{\mu m}{px}$
7	49	melanoma	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23 \frac{\mu m}{px}$